

Impact Statement: Agriculture and Climate Domain-Specific SLM

1. Project Purpose

The purpose of this project is to develop a domain-specific Small Language Model (SLM) capable of generating coherent and contextually relevant responses to agriculture and climate-related questions.

The project explores how lightweight AI models can be adapted to specialized domains using curated knowledge and supervised training data.

The long-term objective is to investigate how such systems could support access to agricultural and climate information, particularly in contexts where access to specialized knowledge may be limited.

2. Potential Positive Impact

A well-designed agriculture and climate SLM could provide several benefits.

Improved Access to Information

The system could make agricultural and climate-related information easier to access through natural language questions.

Users could potentially ask questions about topics such as:

- crop management;
- soil health;
- water management;
- climate adaptation;
- agricultural practices.

Support for Knowledge Access

The model may help organize and present existing domain knowledge in a more accessible conversational format.

Rather than replacing agricultural experts, the system could serve as an additional tool for accessing relevant information.

Lightweight AI for Resource-Constrained Environments

Small Language Models require fewer computational resources than very large models.

This makes them particularly interesting for experimentation in environments with limited computing infrastructure and could support the development of more accessible AI solutions.

Research and Innovation

The project contributes practical experience in:

- domain-specific AI;
- dataset preparation;
- instruction tuning;
- parameter-efficient fine-tuning;
- evaluation of small language models.

3. Potential Risks

Despite its potential benefits, the system also presents important risks.

Incorrect or Misleading Advice

Language models can generate incorrect or unsupported information. In agriculture, incorrect recommendations could potentially affect crop productivity, resource use, or decision-making.

Over-Reliance on AI

Users may treat generated answers as professional advice. The system should therefore not be presented as a replacement for agronomists, climate experts, or local agricultural extension services.

Limited Geographic Context

Agricultural practices depend heavily on local conditions. A recommendation may vary depending on:

- climate;
- soil type;
- crop variety;
- season;
- water availability;
- local farming practices.

The model may not always capture all these factors.

Bias and Data Limitations

The model can inherit limitations or biases from the training data. If some regions, crops, or agricultural practices are underrepresented, model performance may be uneven.

4. Risk Mitigation

To reduce potential risks, this project adopts several principles:

- use domain-specific contextual information;
- preserve metadata such as crop, zone, and topic;
- evaluate generated responses;
- clearly communicate model limitations;
- avoid presenting generated answers as guaranteed expert advice;
- encourage verification of important recommendations with qualified professionals.

5. Human-Centered Use

The intended role of the system is to support—not replace—human expertise.

Agricultural professionals, researchers, farmers, and local extension services remain essential for interpreting recommendations within real-world environmental and social contexts.

Human feedback should be incorporated into future versions of the system to improve reliability and relevance.

6. Long-Term Vision

Future development could include:

- larger and more diverse datasets;
- expert-reviewed training examples;
- improved retrieval of relevant agricultural knowledge;
- multilingual support;
- localization for specific regions and communities;
- stronger evaluation with domain experts.

The ultimate goal is to explore responsible and practical ways of applying AI to agriculture and climate challenges while recognizing the importance of human expertise and local knowledge.